

Lost in the Pipeline: Investigating Nitrous Oxide Waste from Undetectable Leaks at an Academic Hospital

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Background

- Nitrous oxide (N₂O) is a potent greenhouse gas with a global warming potential (GWP100) 273x that of CO₂ and an atmospheric lifespan of 114 years.
- By comparison:

Anesthetic Agent	GWP100	Atmospheric Lifespan
Nitrous Oxide (N ₂ O)	273	114 years
Sevoflurane	~130	~1.1 years
Isoflurane	~510	~3.2 years
Desflurane	~2,540	~14 years

- N₂O contributes up to 80% of anesthetic gas-related emissions in hospitals.
- Global studies show 47–99.8% of N₂O purchased is wasted, with some hospitals reporting 95% loss.
- At the **University of Miami Hospital**, a known but undetectable leak in the central manifold raised concerns about sustainability and system design.

Objectives

- Quantify** N₂O waste at Uhealth , a large academic center.
- Confirm** that the leak is not occurring in patient care areas through clinical zone monitoring - VaporTrak.
- Assess** the environmental and financial impact of this waste.
- Explore** sustainable alternatives to central N₂O delivery systems.

Methods

- Passive dosimeter gas monitoring (VaporTrak) conducted for 8 hours per OSHA/NIOSH protocol in ORs, PACU, ambulatory, and endoscopy suites.
- N₂O administration data pulled from Epic anesthesia records and compared to purchase logs.
- Environmental impact calculated using an EPA GHG emissions tool (CO₂-equivalent emissions).
- Visual confirmation of cylinder returns: 8 H-cylinders/month found to be completely empty.

Table. Monthly nitrous oxide usage: patient-administered volume and waste percentage based on total procurement (115,592L)		
Month	Patient Administered Volume (L)	Waste %
Aug-2023	8,610.80	92.55
Sep-2023	13,061.60	88.70
Oct-2023	7,598.40	93.43
Nov-2023	11,259.20	90.26
Dec-2023	4,249.50	96.32
Jan-2024	7,461.20	93.55
Feb-2024	11,012.40	90.47
Mar-2024	6,290.70	94.56
Apr-2024	7,387.40	93.61
May-2024	12,328.30	89.33
Jun-2024	4,632.20	95.99
Jul-2024	7,031.80	93.92
Aug-2024	7,117.30	93.84
Sep-2024	7,522.60	93.49
Oct-2024	10,059.30	91.30
Nov-2024	8,671.80	92.50
Dec-2024	6,013.30	94.80
Jan-2025	8,168.40	92.93
Feb-2025	6,476.50	94.39
Mar-2025	3,790.60	96.72
Apr-2025	2,755.50	97.62
May-2025	4,250.60	96.32
Jun-2025	3,130.30	97.29



Figure 1: 16-tank nitrous oxide manifold at our institution. The system alternates monthly between two banks of eight tanks, and is controlled by an automated sensor between banks to ensure continuous supply.

Results

- No N₂O detected in clinical areas (detection limit: 0.56 ppm) → confirms leak is in the central manifold/piping infrastructure in unattainable areas of the hospital.
- Of 115,592 L of N₂O purchased monthly, only an average 7,400 L (6.4%) is administered to patients.
→ 92.8% of N₂O is lost.
- This waste equals **60,572.4 kg of CO₂e/month**, comparable to:

Environmental Impact of Monthly Nitrous Oxide Waste	
Emission Equivalent	Monthly Value
CO ₂ e from N ₂ O Waste (kg/month)	60,500.00
Equivalent Gasoline Burned (gallons)	~6,800
Equivalent Car Miles Driven	~154,000
Incandescent bulbs switched to LED for 10 years	~2,300
Number of smartphones charged	~4.8 million

Conclusion

- Despite low financial cost (~\$792/month), environmental impact of N₂O waste is quantifiable.
- Current piped N₂O infrastructure leads to inevitable systemic loss.
- Gas monitoring confirms **leaks are not in clinical zones**—making detection and repair difficult.
- Replacing piped systems with E-cylinders is a sustainable and evidence-based solution.

Future Directions

- Implement a pilot program using E-cylinder-only nitrous delivery in a limited number of ORs.
- Evaluate the pilot for:
 - Cost-effectiveness
 - Clinical usability and workflow impact
 - Reduction in N₂O waste and emissions
- Use findings to inform a data-driven decision on whether to pursue widespread decommissioning of the central N₂O pipeline system.

References

